KEYNOTE

AI agents in drug discovery: applications and case studies

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AI agents are emerging as transformative tools in drug discovery, with the ability to autonomously reason, act and learn through complicated research workflows. Building on large language models and specialized tools, these systems can integrate biomedical data, execute tasks, conduct experiments and iteratively refine hypotheses. We provide a conceptual overview of agentic AI architectures and illustrate their applications across key stages of drug discovery, including literature synthesis, automated protocol generation, toxicity prediction, small-molecule synthesis, drug repurposing and end-to-end decision-making. Early implementations demonstrate substantial gains in speed, reproducibility and scalability.

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Equal contribution.

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We discuss the challenges related to data heterogeneity, system reliability, privacy, benchmarking and outline future directions toward technology in support of science and translation.

Keywords: Drug discovery; AI agent; agentic AI; autonomous science; data-driven; AI-driven; robotic; large language model; autonomous laboratories

Introduction

Drug discovery is a long, expensive and high-risk process that spans from identifying new biological targets to delivering safe and effective medicines to patients.[p1] At a general and simplified level, it follows a sequence of connected stages: target identification and validation, hit discovery, lead optimization through iterative design-make-test-analyze (DMTA) cycles, preclinical safety assessment and, finally, clinical development.[p2],[p3],[p4],[p5] Each stage produces large amounts of data, ranging from omics and imaging to assay readouts, pharmacokinetic studies and clinical outcomes.[p6] Despite technological advances, such as non-animal new approach methodologies,[p7] the use of machine learning (ML) models and laboratory automation, complex decision-making at key junctures is still largely manual, relying on cross-disciplinary teams to search, collate and interpret scattered evidence.[p8] This reliance on fragmented workflows contributes significantly to the rising cost, longer timelines and high attrition rates that characterize drug R&D. In addition, many individual steps of drug discovery remain tedious, slow and over-reliant on human memory and multidisciplinarity.

Computational methods, and particularly advacned AI approaches, have already influenced methods used in drug discovery in multiple ways. Predictive AI in the form of ML has enabled QSAR models that predict compound properties to be lifted to a new quantitative and likely qualitative level.[p9],[p10],[p11] Generative AI has made it possible to design new chemical structures with desired properties, and large language models (LLMs) are starting to structure machine-readable data to propose hypotheses[p12],[p13],[p14],[p15],[p16] (Figure 1a). Still, these paradigms share a common limitation: they comprise passive implementations requiring human actions for repetitive tasks such as to prepare inputs and to interpret outputs.[p16],[p17] They cannot autonomously orchestrate the multi-step reasoning and decision making required across the complex, interdependent processes of drug discovery.

A new wave of systems, collectively termed agentic AI, offers the potential to overcome these limitations (Figure 1a). Agentic AI builds on the recent advancements in LLM reasoning capabilities but couples them with external tools, memory and data sources, enabling systems that can 'think', 'act', 'observe' and 'reflect' in iterative loops. Rather than serving as a single predictive or generative model, these agents function as adaptive independent components that can act together in retrieving literature, triaging compounds, predicting toxicity endpoints such as liver- or cardio-toxicity,[p18] planning experiments and even interfacing with lab automation.[p19] This agentic paradigm builds on recent demonstrations of autonomous scientific discovery, including systems that autonomously plan and execute chemical experiments[p20] and multi-agent platforms for biomedical research coordination,[p21] suggesting that the integration of reasoning, tool use and iterative refinement could fundamentally reshape how drug discovery is conducted. However, at the same time, to achieve progress in the real world, we need technical progress to be matched by improvements in our ability to specify a particular problem as a 'task' to a ML model – a problem not only present in the context of drug discovery.[p22] This overview positions agentic AI directly within the drug discovery pipeline. We first outline the conceptual foundations and system architectures of agentic AI. We then examine applications across key stages of drug R&D, from target discovery and DMTA cycles to preclinical safety and drug repurposing, highlighting case studies where agents are already beginning to reshape practice, as part of the AIAgents4Science Consortium (broad.io/aiagents4science). Finally, we discuss important challenges, including data heterogeneity, infrastructure integration and system reliability, and propose future directions.

Agentic AI: concepts and system design principles

The new AI application wave for drug discovery

Recent advances in predictive and generative AI, especially LLMs, have paved the way for the evolution of agentic AI in drug discovery. This shift builds on engineering efforts to coordinate predictive and generative models, allowing AI systems to autonomously execute a range of drug discovery tasks. At the heart of these systems are LLMs with 'reasoning' capacity,[p5],[p19],[p21],[p23],[p24] the extent of which remains debated.[p25],[p26] A stand-alone LLM is a generative model that is good at generating outputs, such as text, SMILES strings and protein sequences. However, its capabilities are limited to generation; it cannot interact with the real world to perform meaningful actions[p27] or access external data beyond its training set, whereas drug discovery research requires capabilities beyond generating objects. Decision-making in drug discovery requires the ability to search biological databases and the web for information on targets, novelty and clinical trials, and to analyze and integrate complex biomedical data such as single-cell datasets, assay readouts and ML model outputs. Furthermore, the ability to implement and interpret experiments is key to drug discovery.

To acquire such capabilities, the LLM must integrate with external tools.[p20],[p21],[p23] Here, tools refer to any external components integrated with an LLM that can be called to perform specific tasks. In the scope of drug discovery, we categorized agent tools into four common types based on their functionality (Figure 1b). (i) Perception tools serve as the system's augment layer, gathering information from structured and unstructured biomedical databases, such as ChEMBL, PubChem, STRING and Reactome.[p28],[p29],[p30],[p31] (ii) Computation tools enable predictions, simulations, data analysis or other computational actions in the drug discovery field. These tools often serve as wrappers for pre-trained models, such as AlphaFold,[p32] or data processing pipelines, like NextFlow.[p33] (iii) Action tools provide

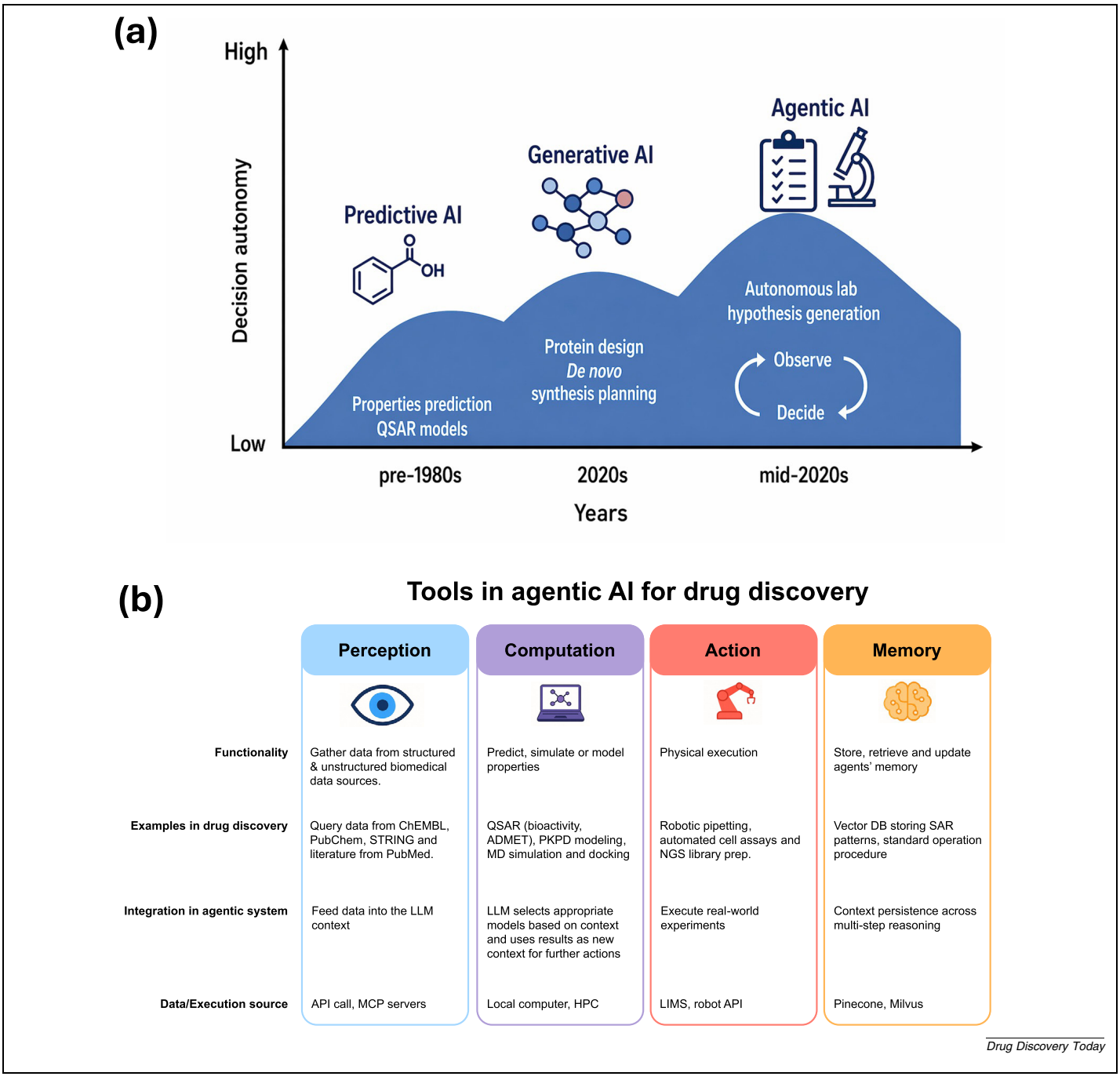


FIGURE 1

(a) The three evolutionary waves of AI: from predictive models to generative and agentic systems. **(b)** Four typical tool types in agentic AI for drug discovery. Perception tools enable data collection and integration, computation tools perform analytical and predictive modeling, action tools execute real-world experimental operations and memory tools manage knowledge storage, retrieval and continuous learning.

the agentic system with the ability to act in the real world. For example, the agentic system connects to robotic pipetting, automated cell based assays or next generation sequencing library preparation to perform physical execution. These action tools robustly close the loop between *in silico* design and empirical validation. (iv) Memory tools maintain persistence across tasks and sessions by storing, retrieving, compressing and updating the agent's working knowledge. In drug discovery contexts, memory databases might capture SAR patterns, accumulated toxicity findings and/or other high value knowledge that are repeatedly reused. This memory layer provides context persistence across multi step reasoning, which enables the system to refine hypotheses over time.

Together, these four types of tools enable agentic AI to implement end to end pipelines in drug discovery. Perception receives multimodal evidence, such as omics datasets, structural biology data, literature, patents and clinical trial records, to generate and refine target or mechanism hypotheses. Computation

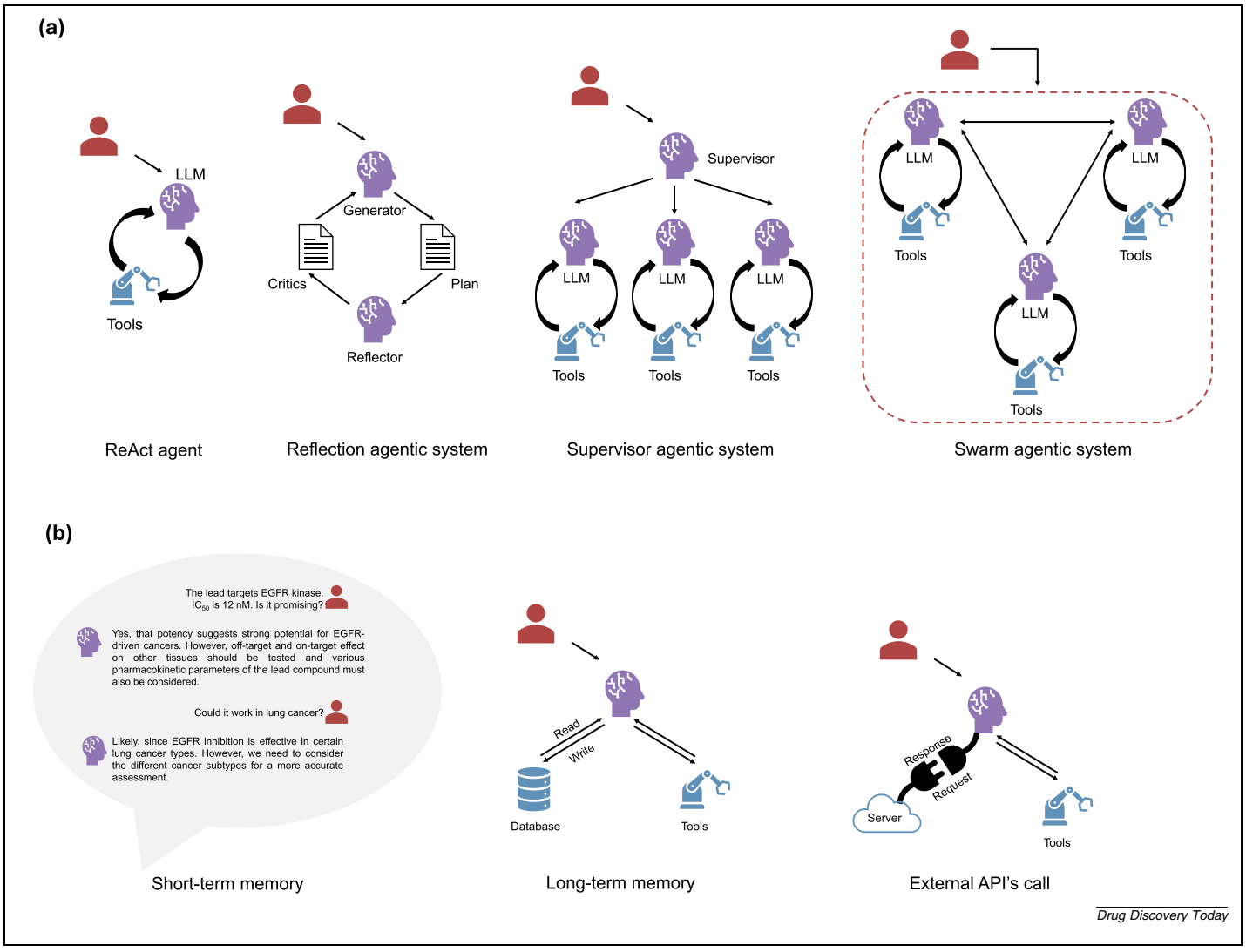


FIGURE 2

(a) Architectures of agentic AI systems. ReAct agent: a reasoning-acting loop where a large language model (LLM) dynamically selects and invokes tools. Reflection agentic system: a multi-LLM setup in which a generator produces plans, a reflector critiques them and iterations improve strategy. Supervisor agentic system: a hierarchical multi-agent architecture with a supervisor LLM delegating subtasks to specialized LLM agents, each capable of reasoning and tool use. Swarm agentic system: a decentralized multi-agent system with all agents connecting to each other, each capable of reasoning and tool use. **(b)** Memory and external integration in agentic AI systems. Short-term memory: the LLM maintains conversational context within the current interaction. Long-term memory: the LLM can read from and write to an external database, allowing knowledge persistence across conversations. External application programming interface (API) calls: the LLM interacts with third-party APIs to fetch real-time data.

translates these hypotheses into quantitative predictions, including target tractability assessment, structure-based design, molecular property prediction, ADMET profiling and biomarker identification. Action tests these predictions through experimental design and execution, such as compound synthesis prioritization, HTS, CRISPR perturbation studies or *in vitro*/*in vivo* validation. Memory captures experimental outcomes, negative results and contextual metadata to continuously update the discovery context, improve decision policies and guide subsequent DMTA cycles.

Architectures

To better understand the capabilities of agentic AI, it is helpful to examine several established system architectures. The most basic agentic architecture is ReAct, which stands for reasoning-acting[(p34)] (Figure 2a). This design enables the LLM to dynamically select and execute the necessary tools when receiving tasks. This architecture also enables an iterative loop between the reasoning and acting processes.[(p34)] The LLM actively determines when the process will terminate with minimum human intervention. This enhances the flexibility and adaptability of the agentic system, making it more akin to the style of research that involves thinking critically. The iterative loop process mimics the DMTA cycle in drug discovery, strengthening the hypothesis with each iteration.

Another architecture is the reflection agent (Figure 2a), in which multiple LLMs are connected, enabling them to communicate with each other and critique one another's reason-

ing.[p35] This system mimics working in pairs situations, where medicinal chemists sit together to discuss the synthesis plan for compounds, or software engineers program as a pair. Therefore, this architecture is especially effective for tasks that require discussion and strategic planning. For example, it can be applied to plan multi-step synthesis routes or design experimental workflows for HTS.

Multi-agent architectures are also gaining traction, with the emergence of supervisor architectures (Figure 2a). This architecture is modeled on the way a research team is organized. We have one supervisor agent who focuses on task delegation, and many expert agents who deeply reason and act on a relevant specific task.[p19],[p36] When receiving a task, the supervisor decomposes it into sub-tasks that are suitable for each specialized agent. Then, the supervisor delegates tasks to the sub-agents to execute and collect results. This process also enables an iterative loop similar to the ReAct agent, which makes the supervisor architecture goal-oriented. However, in a supervisor architecture, every communication must go through the supervisor because specialized sub-agents only connect to the supervisor, not to each other. This repeated communication with the supervisor lengthens the process's context, which is a limitation for LLM with restricted context-window length. As a solution for this issue, the swarm architecture (Figure 2a) aims to decentralize the role of the supervisor agent, with each agent connecting to every other agent, facilitating the collaboration between them.[p37] Additionally, agents can exchange data and insights through communication protocols. IBM's Agent Communication Protocol (ACP)[p38] and Google's Agent2Agent (A2A),[p39] which recently merged under the Linux Foundation, aim to create a standardized interface for agents to exchange data independently of the underlying framework or location which enables agents to collaborate across departments of the same organization and across companies and labs all over the world. Overall, these designs represent just the beginning of a rapidly evolving field. As we continue to explore new effective agentic architectures, many novel and domain-specific suggestions will emerge, tailored for different problems in drug discovery. The design space is wide open – from closely coupled experiment–agent systems to entirely autonomous research platforms.

Memory: effective components for agents to learn

Despite the autonomous potential of AI agents, drug discovery remains a long-running process involving multiple stages and stakeholders. As a result, an effective memory layer for AI agents becomes crucial for translating this technology into real-world applications. An agent's memory gives it the ability to learn from experience, maintain coherent interactions and accomplish complex tasks. There are two types of agent memories: (i) short term, which reflects the working space for immediate context; and (ii) long term, for persistent knowledge. In drug discovery, an agent's memory allows it to integrate continuously generated data. For instance, the agent can connect a finding from a new experimental result with internal memory of scientific publications to propose a novel hypothesis. It can also refine structural features of a new compound based on hurdles faced in past discovery programs. This active synthesis can allow an AI agent to identify novel hypotheses based on past findings and adapt strategies in real-time.

Agent short-term memory

AI agents rely on short-term memory to process immediate information, similar to working memory in humans. This buffer, also called the context window, is finite and holds the agent's recent conversational history along with data from application programming interface (API) responses and files to inform its actions (Figure 2b). This mechanism enables in-context learning (ICL), where the agent uses the provided context to generate responses without altering its underlying parameters.[p40],[p41]

Agent long-term memory

Drug discovery programs span years and generate heterogeneous data across multiple DMTA cycles. Effective agentic systems therefore require persistent long-term memory that captures biological context, experimental outcomes and decision rationales. To retain information permanently, agents use long-term memory, both internal and external. Internal long-term memory is the parametric knowledge encoded within the neural network's weights during its initial training. In drug discovery, this includes foundational chemical and biological knowledge such as protein–ligand interaction patterns,[p42] medicinal chemistry rules,[p43],[p44] ADMET principles[p45] and common assay artifacts.[p46] This internal knowledge can be updated through several methods such as periodically retraining the model on new, domain-specific data to integrate new foundational knowledge (continued pre-training),[p47] adapting the model for specific tasks, styles or instruction-following capabilities (fine-tuning)[p48]; or combining the weights of multiple specialized models to create a single, more knowledgeable agent, which is useful when training data is proprietary (model merging).[p49] However, internal parametric memory is costly to update and does not easily capture rapidly evolving experimental results.

External long-term memory (Figure 2b) provides a dynamic and persistent storage solution, most commonly implemented through retrieval-augmented generation (RAG) systems. In a typical RAG architecture, external documents, scientific publications or research data are embedded into numerical representations and stored in a vector database. This foundational approach has inspired more-advanced techniques. For instance, AgenticRAG[p50] employs an agent that can iteratively refine and reformulate search queries, improving retrieval accuracy. Similarly, GraphRAG[p51] organizes knowledge into structured graphs and retrieves information by traversing entities and their relationships, rather than relying solely on text chunks. Julien et al. have shown that GraphRAG can surface rare but important associations, such as identifying novel drug-gene-disease links for conditions like asthma and Alzheimer's disease, outperforming classical RAG.[p52]

Beyond the long-term memory

In drug discovery, static long-term memory, whether parametric or RAG-based, is insufficient because biological knowledge, clinical trial evidence and experimental data evolve continuously. Thus, AI agents need to extend memory through active interaction with live data sources, analytical tools and laboratory systems. Agents use a feature known as tool calling (Figure 2b) to interact with external databases and services via APIs; this allows

them to query specialized databases like ChEMBL[p28] for chemical data or PubMed[p53] for scientific literature, incorporating the results into their short-term memory. Model context protocol (MCP)[p54] is an emerging protocol standard developed by Anthropic, designed to create a universal interface between agents and external data sources.

Accelerating drug discovery with agentic AI: areas of application

Here, we present several case studies in which (multi-)agentic systems were developed to focus on specific functional areas of drug discovery and development. Table S1 (see supplementary material online) summarizes the domains, architectural designs of the agentic systems, key used tools and notable outcomes for each case study. We acknowledge that several case studies presented in this industry overview involve proprietary industrial systems where complete source code and training data cannot be disclosed. To mitigate this limitation, we report specific algorithmic parameters, quantitative outputs and benchmark comparisons against public baselines, enabling readers to evaluate the evidentiary basis of each claim independently of access to the underlying codebases.

Comprehensive literature analysis for molecular prioritization

Comprehensive literature analysis is a tedious and time-consuming task in preclinical drug discovery. Medicinal chemists must search patents and publications for structurally similar analogs, extract SAR data, cross-reference measures and synthesize insights. This process typically requires weeks, delaying prioritization and synthesis, especially when rapid iteration is needed early in discovery. A multi-agent framework could transform this process by autonomously coordinating literature analysis across distributed data sources (Figure 3). A system from Coincidence Labs uses a hierarchical supervisor pattern – an orchestrator agent manages specialized sub-agents with distinct domain

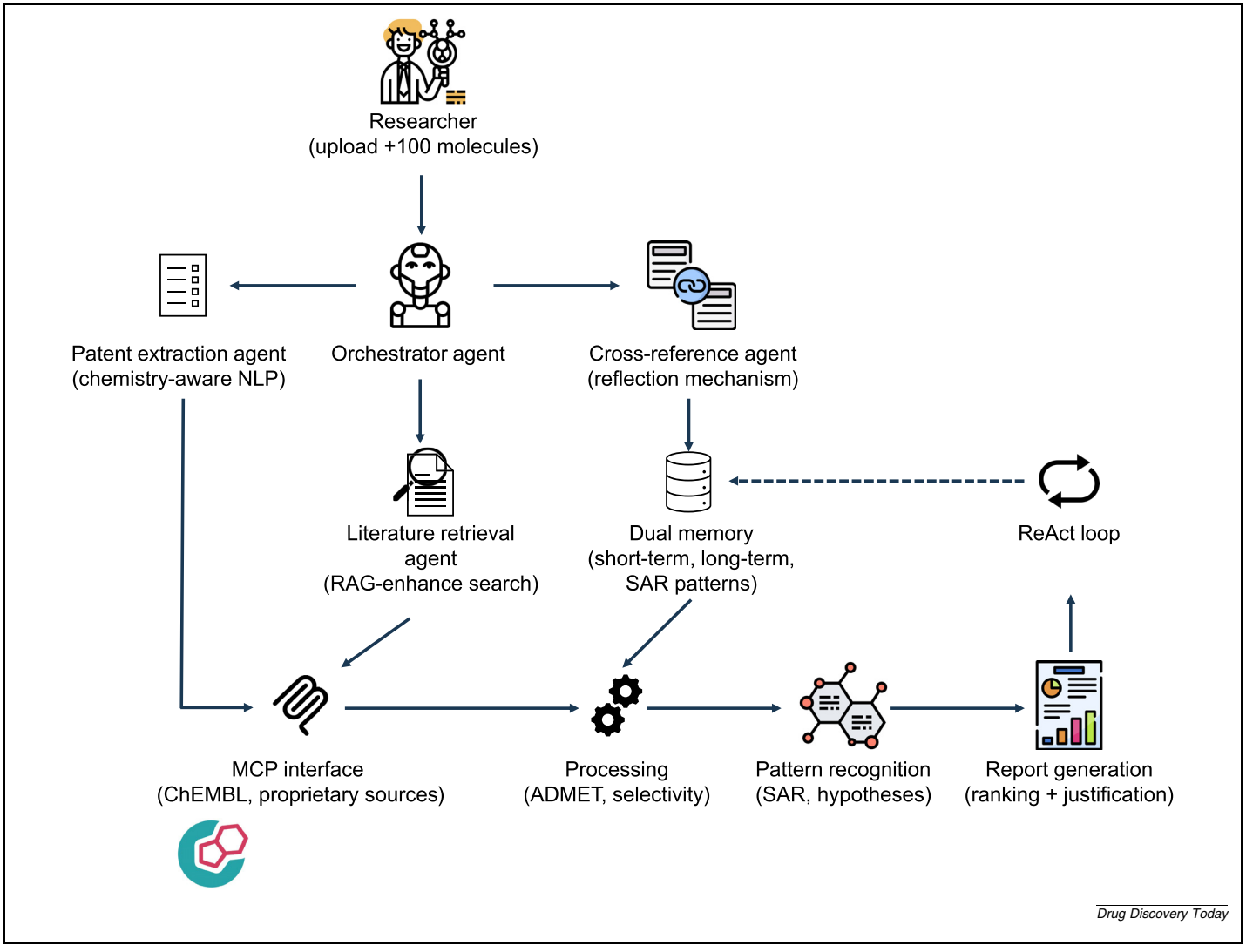


FIGURE 3

Agentic system design for comprehensive literature analysis in connection to molecular structures and/or small molecules in early preclinical drug discovery. The system's center is an orchestrator agent that harmonizes many different components, including patent extraction agent, cross-reference agent, literature retrieval agent and dual memory, with molecular structure and data at the heart.

expertise. Patent extraction agents retrieve patents and extract SAR data using chemistry-aware parsing and RAG-enhanced retrieval; literature retrieval agents gather ADMET properties via semantic search; and cross-reference agents identify conflicting data using comparative analyses. An MCP standardizes database interactions with ChEMBL, PubChem and patent repositories. The RAG pipeline ensures each data point is traceable to its source. A dual-memory architecture drives continuous improvement: short-term memory stores session findings; long-term memory retains scaffold-property patterns and query strategies, enabling recognition of previously analyzed series across projects.

A typical workflow starts with uploading hundreds of molecular structures. The system autonomously: (i) retrieves structurally similar analogs from patent and literature databases using similarity search (Tanimoto similarity >0.7, substructure matching); (ii) extracts SAR data from associated documents; (iii) generates ADMET predictions for novel compounds using models trained on internal and public datasets (covering solubility, permeability, CYP450 liability and hERG cardiotoxicity); and (iv) aggregates findings into structured reports with citation provenance. When sources report conflicting data (e.g., varying IC_{50} values for the same reference compound), cross-reference agents flag discrepancies and present all variants with experimental metadata, such as assay format, cell line and measurement conditions. Source metadata are provided to contextualize retrieved information. Crucially, the RAG pipeline tracks each data point to its original source for verification and reliability.

As a case study, Coincidence Labs used this framework in a Bruton's tyrosine kinase (BTK) inhibitor discovery workflow. The orchestrator agent coordinated parallel operations across sub-agents to assess acalabrutinib's selectivity profile versus novel structural analogs. Acalabrutinib, a second-generation BTK inhibitor for B-cell malignancies, was designed for improved selectivity over first-generation drugs, for example ibrutinib. Patent analysis agents used similarity search on Morgan fingerprints (radius = 2, n-bits = 2048, Tanimoto similarity >0.7) to find patents, literature retrieval agents queried scientific databases for kinase selectivity data (Table 1) and ADMET properties, and cross-reference agents identified conflicting measurements across sources (Table S2, see supplementary material online). When the cross-reference agent encounters multiple potency values for the same compound–target pair, it flags discrepancies exceeding threefold and initiates a structured conflict–resolution workflow. First, the agent retrieves assay metadata from ChEMBL for each measurement (assay format, substrate, preincubation time, measurement type). Second, it classifies each value by category: enzymatic IC_{50}, binding K_d, cellular IC_{50} or occupancy-based measurement. Third, it groups concordant values by assay platform and annotates each entry with its experimental context. Finally, it generates a root-cause annotation explaining the discrepancy (Table S2, see supplementary material online). The analyst reviews the agent's output before incorporation into the final table but the detection, classification and characterization of conflicts is performed by the agent autonomously. The system delivers comprehensive reports with structured molecular tables, extracted SAR data with patent citations, conflict flags, ADMET prediction profiles and complete citation chains. Overall, this agentic workflow allows researchers to focus on interpretation and prioritization instead of manual literature aggregation.

In silico toxicity predictions

The pharmaceutical and chemical industry faces bottlenecks in toxicological assessments. An agentic AI system can greatly enhance chemical toxicity prediction by combining predictive toxicology modeling with cheminformatics to enable safer chemical design. To address this, Human Chemical developed an agentic AI system using the ReAct architecture for *in silico* toxicology. It integrates predictive models and cheminformatics tools, summarizing predicted values alongside structured overviews of scientific literature, regulatory assessments and other data. After each cycle, users can ask follow-up questions or assign additional tasks, making this a human-in-the-loop system that increases research productivity while leveraging necessary human expertise for chemical safety assessment. For example, the team used this agent to assess the endocrine disruption risk of cashmeran, a polycyclic musk used in perfumes. After querying the agentic AI system on cashmeran using its CAS number (33704-61-9), the system automatically initiated a series of toxicological property predictions using proprietary models. The *in silico* toxicity predictions were generated using Human Chemical's bioactivity predictor – an ensemble of multi-task graph neural network (GNN) models that operates directly on molecular SMILES representations. The architecture processes node-level (atomic), edge-level (bond) and graph-level (molecular) features

TABLE 1

Multi-source IC_{50} data extraction: acalabrutinib and multiple kinase targets.

Target kinase	Biochemical IC_{50} (nM)	Cellular IC_{50} (nM)	Inter-assay range	Assay type (cellular)
BTK	3.0–5.1	<10	Low variability	Z'-LYTE, IMAP
TEC	9.7 ± 2.6[(p96)] 37–126 (other studies)	>1000	13-Fold variability	Platelet phosphorylation
BMX	46–598	N/A	Tenfold variability	N/A
EGFR	>1000–3513	>10 000	Consistent	A431 EGF-induced
ERBB2	>1000	>1000	Consistent	Z′-LYTE
ERBB4	16–140	N/A	Ninefold variability	N/A
ITK	>1000–30000	>1000	Low above threshold	T-cell assays
JAK3	>1000	N/A	N/A	N/A

to learn SAR without reliance on structural alerts or lookup tables, enabling predictions for novel chemical structures outside curated rule sets.

The training dataset was developed in close collaboration with advisors at the National Institute of Environmental Health Sciences (NIEHS) and other US Federal Government agencies to ensure data quality and endpoint relevance. This partnership informed the curation strategy for each of the endpoints including the selection of authoritative data sources, the definition of positive and negative class labels grounded in regulatory frameworks, and the application of quality filters to exclude ambiguous or low-confidence study records. This expert-guided curation process was crucial for producing training sets that reflect the biological and regulatory meaning of each endpoint rather than artifacts of heterogeneous source data.

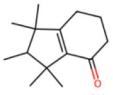
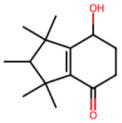
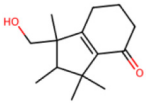
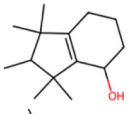
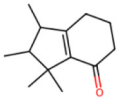
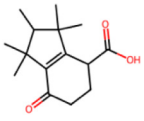
The model is trained in a multi-task framework, jointly predicting all endpoints simultaneously rather than fitting independent single-task models. This design leverages shared molecular representations across related toxicological tasks, enabling the network to learn transferable structural features that generalize across endpoints. Multi-task learning is particularly advantageous in molecular property prediction, where individual endpoint datasets are often small and imbalanced; shared representation learning acts as an implicit regularizer, reducing overfitting and improving performance on data-sparse endpoints by borrowing signals from related tasks. Finally, the system retrieved, synthesized and returned a comprehensive report on the known properties of cashmeran, drawing on the scientific literature and authoritative regulatory statements. The system flagged a potential endocrine disruption hazard based on molecular structure. Recognizing that risk is a function of intrinsic hazard and systemic exposure, the agentic system pursued two complementary lines of inquiry: refined hazard characterization and exposure assessment, to determine whether this structural alert translates to meaningful risk.

For hazard refinement, the system retrieved predicted metabolites using BioTransformer 3.0 and evaluated their individual structural hazards, revealing a progressive reduction in predicted endocrine bioactivity across successive metabolic steps compared to the parent compound (Table 2; and Figure S1, see supplementary material online). Structure–activity analysis indicated that the terminal metabolite of cashmeran, a carboxylic acid, is predicted to have low intrinsic endocrine hazard owing to high polarity and poor membrane permeation. For exposure characterization, the system identified that cashmeran undergoes rapid metabolic clearance, which limits systemic exposure to the parent compound, the species of greatest predicted hazard concern.

Simultaneously, the agentic AI system conducted a comprehensive scientific literature review of cashmeran, identifying evidence supporting low intrinsic hazard and low systemic exposure. On the hazard side, the system found: published *in vitro* HTS data from ToxCast, which found cashmeran to be inactive in estrogen-related assays at non-cytotoxic concentrations[p55]; a lack of adverse endocrine-related effects reported in a 90-day repeated gavage dosing study of cashmeran; and a classification of 'not toxic' by the Australian Industrial Chemicals Introduction Scheme. On the exposure side, the system cited: an *in vitro* observation of notably rapid cashmeran metabolism relative to eight other fragrance chemicals, with faster metabolic rates in humans compared with trout (half-life <1 h in humans

TABLE 2

Hazard prediction results of cashmeran metabolites.

Metabolite	Structure Description	Structure	Endocrine Activity Probability	Confidence Score	Comparison to Parent
Parent Cashmeran	Original compound		73.7%	1.0	–
Aromatic Hydroxylated	Hydroxyl group added to aromatic ring		61.6%	0.5	↓ 12.1%
Aliphatic Hydroxylated	Hydroxyl group added to methyl side chain		60.7%	0.75	↓ 13.0%
Ketone Reduced	Ketone reduced to alcohol		61.7%	0.875	↓ 12.0%
Demethylated	One methyl group removed		70.2%	1.0	↓ 3.5%
Carboxylic Acid	Oxidation to carboxylic acid		45.6%	0.125	↓ 28.1%

versus <1 day in trout),[p56] which drastically limits systemic exposure to the parent compound; a relatively high NOAEL of 1875 mg/kg/day (oral, rat) established in a reproductive toxicity assay,[p57] indicating that adverse effects are absent even at high systemic doses; and an ECHA report of 'no adverse effect observed' regarding toxicity to reproduction.[p58]

Integrating both lines of evidence, low intrinsic hazard of metabolites combined with minimal systemic exposure to the parent, the system assessed overall endocrine disruption risk as low. When comparing *in silico* predictions with the literature, the system found strong agreement across hazard and exposure dimensions. It also noted that simple *in vitro* hazard assays do not always reflect real-world risk because they lack a competent metabolic compartment to simulate toxicokinetics (TK). By bridging structural hazard alerts with metabolic clearance data, the agentic AI prevents safe chemicals from being falsely abandoned on the basis of hazard alone. These results demonstrate the utility of agentic AI in chemical safety assessment, especially when complex, multi-step analyses spanning hazard characterization and exposure assessment are required. Such systems are applicable to drug discovery, cosmetics, industrial chemistry, materials science and environmental toxicology.

Automating protocol design and execution

Designing and validating assays to support drug discovery is a slow, expertise-intensive process. Scientists must review extensive literature, compare existing assay designs, select suitable reagents or design primers and probes, and adapt protocols to specific experimental platforms and automation systems. For assays intended to provide evidence or as a biomarker, regulatory considerations further increase complexity. Documentation, experimental controls and reporting must align with regulatory expectations and guidelines (e.g., ICH and FDA) for assay performance, reproducibility and traceability. These requirements mean that validation (i.e., establishing sensitivity, specificity, dynamic range and robustness across instruments and reagents) often requires multiple experimental iterations and cross-functional review – a process that can take months and slow drug development timelines. Traditional workflows further fragment this process across disconnected stages, including literature review, protocol drafting and coding for automation, thereby increasing coordination time, cost and variability across labs.

Agentic AI can address these inefficiencies by coordinating the entire assay-development workflow in a closed-loop cycle: searching literature, reasoning about experimental design, generating protocols, producing executable lab instructions and iteratively refining experiments based on results. With tools for literature retrieval, comparative assessment and robotic code generation, agents can autonomously design and refine assays. For example, an agent can retrieve and evaluate published qPCR methods, identify those with the highest sensitivity and specificity, generate a MIQE-aligned (Minimum Information for Publication of Quantitative Real-Time PCR Experiments) protocol and translate it into executable code for automation platforms. By connecting these steps through a closed-loop process, the system reduces human effort while improving reproducibility and traceability.

As an example, authors affiliated with Potato developed a multi-agent system: Tater, which integrates RAG with specialized scientific tools spanning planning, execution and iteration. A scientist provided the following task: 'Derive an automated protocol on an Opentrons to perform qPCR quantification of adeno-associated virus (AAV). This is for preclinical grade material but the assay should be able to be developed into a clinical release test. Search the literature for appropriate methods, identify those with the best sensitivity and specificity that meet FDA requirements and write a detailed protocol including what data analysis needs to be done'.

Within 2 h, Tater generated a curated literature set and citation map of qPCR methods for AAV quantification. It then produced a comparative table of assay parameters (sensitivity, specificity, MIQE alignment, regulatory suitability), a complete protocol with operational details and controls (Figure S2, see supplementary material online) and translated it into executable code for an Opentrons liquid handler. A structured report summarized assay scope, analytical criteria, automation and statistical analysis templates. The agent synthesized information from multiple sources, creating a validated workflow and automation code. Compared to manual design, Tater reduced setup time and human effort by more than two orders of magnitude (Table 3).

Even after downstream verification and testing, efficiency gains are substantial. Tater's outputs, including literature syntheses, protocol drafts and automation scripts, require human review for reagent compatibility, calibration and compliance; but these steps add hours, not weeks. The agent enables a two-orders-of-magnitude reduction in human effort while maintaining traceability. Empirical validation is still necessary to confirm assay performance and regulatory compliance. Rather than replacing scientific judgment, the system shifts human input from manual assembly to higher-value reasoning and validation.

TABLE 3

Automating qPCR assay development using Potato's Tater agent. Manual qPCR design process requires weeks of literature review, manual protocol drafting and scripting by automation engineers. Tater's integrated agentic workflow links literature retrieval, reasoning, protocol generation and automation script output. Example outputs include comparative methods table, MIQE-aligned protocol and Python-based Opentrons code. Quantitative comparison demonstrates compression of the 4-month workflow to <2 h.

Workflow stage	Manual (typical lab estimate)	With Tater	Δ Improvement (based on minimum manual estimate)
Literature review & method comparison	5–14 days	13 min	≈550× faster
Protocol drafting & MIQE validation	6–10 h	8 min	≈45× faster
Translating protocol to script	3 weeks–3 months	55 min	≈260× faster
Total end-to-end design cycle	1–4 months	1 h 39 min	≈400× faster

By autonomously reasoning from literature through to executable automation code, an agentic AI system compresses what used to require multidisciplinary coordination into a single, auditable workflow. A process that traditionally spans months of cross-functional effort was completed in <2 h, representing a >400× reduction in cycle time. This result establishes a quantitative benchmark for next-generation autonomous scientific systems, demonstrating how automation can augment human expertise while preserving scientific integrity and oversight. Although this example reflects the time savings for a single design iteration, in practice most assays require multiple design-test-optimize cycles to achieve acceptable assay performance metrics. Because agentic systems like Potato now extend beyond single iterations to autonomously manage sequential design and optimization cycles, additional time savings and efficiency gains can be realized.

Accelerating drug discovery with Virtual Scientists

Drug discovery workflows are often fragmented across biology, chemistry and clinical domains. Integrating heterogeneous data, tools and models is challenging, and conventional AI systems frequently lose experimental context, treat tasks as isolated and make experiment prioritization difficult while maintaining transparency, reproducibility and scalability. An infrastructure-focused platform powered by Virtual Scientists, a multi-agent AI system, can accelerate drug discovery. At Kiin Bio, Virtual Scientists integrate >100 tools and AI models across biology, chemistry and clinical domains, from API-based tools to large-scale bioinformatics pipelines and GPU-based models. This enables the system to propose, evaluate and prioritize *in silico* and lab-based experiments with transparency, reproducibility and scalability. The Virtual Scientist is a specialized agent designed for a distinct scientific role, such as: (i) literature review, biological data exploration and dataset identification; (ii) running large-scale pipelines and performing multi-omics analyses like transcriptomics, proteomics, variant analysis and polygenic risk scoring; and (iii) molecular design and screening, including virtual screening, protein folding, property prediction and novelty checking. Agents operate autonomously and contribute to a shared organizational memory, supporting coordinated decision making across teams. A core innovation of the Virtual Scientist system is its ability to capture and reason over experimental context. By embedding context into every stage, Virtual Scientists understand each question's connection to prior evidence and project goals. This context awareness enables more-accurate prioritization of experiments, reduces redundancy and increases reproducibility, ultimately accelerating the path to meaningful scientific insight.

As a case study, Kiin Bio supported the development of a preclinical drug discovery program for idiopathic pulmonary fibrosis (IPF). The goal was to identify market opportunities, prioritize novel targets and generate and rank potential small-molecule hits. Virtual Scientists orchestrated this nonlinear workflow (Figure 4), integrating multiple agents to analyze public omics data, literature, RNA-seq analytics, structural models and generative chemistry. Outputs included disease landscape reviews, ranked omics datasets and initial small-molecule candidates with predicted binding affinities. The experiment-level context, defined by users and evolving throughout the process, directly informs next activity suggestions. The workflow could be executed end-

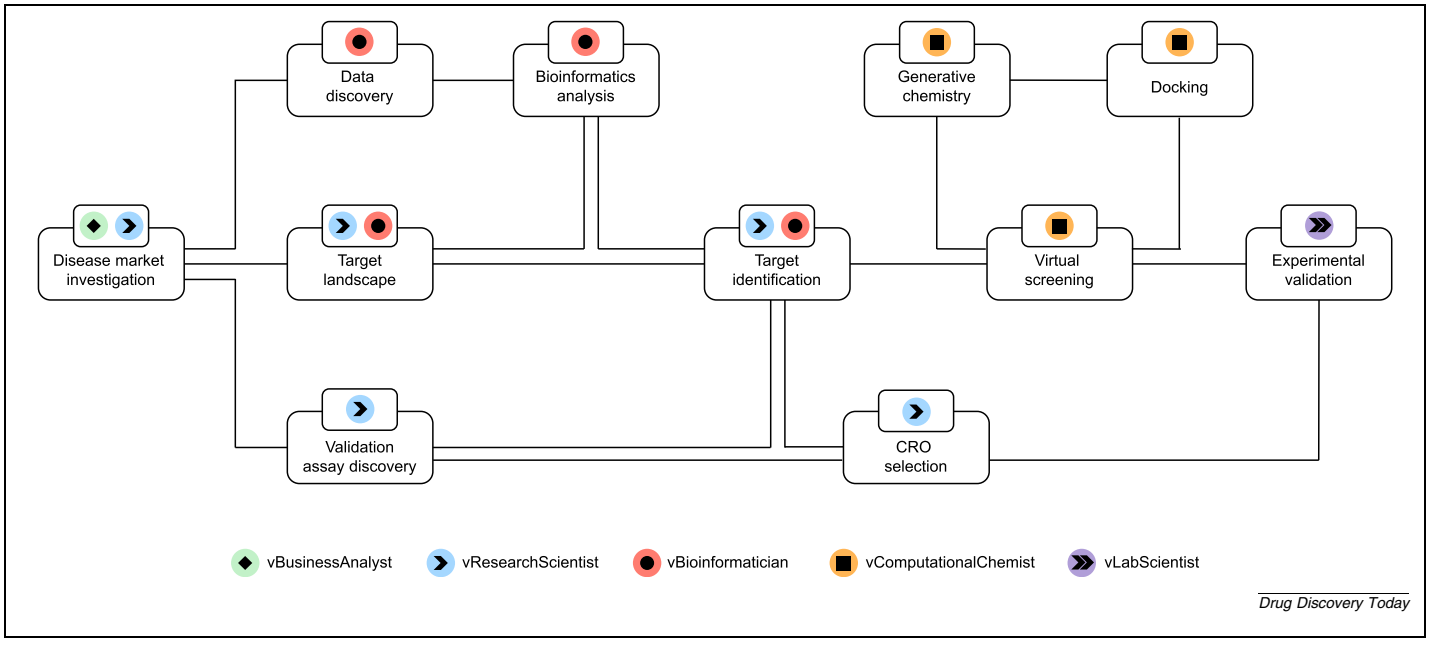


FIGURE 4

Drug discovery workflow enabled by Kiin Bio's Virtual Scientist platform. Each block represents a distinct research plan within a larger experimental workflow. Plans are tagged with the Virtual Scientist agent responsible, illustrating how specialized agents collaborate to execute and coordinate complex end-to-end discovery processes.

to-end in <2 h, compared with the 2–3 week estimated duration when conducted manually across independent systems, and could also be automatically rerun as new data become available. Processes like molecular docking and generative chemistry were streamlined through AI-agent orchestration. Operational tasks such as CRO selection that typically require months were completed more efficiently via automated coordination.

Drug repurposing for rare diseases

Rare diseases are associated with inherently small patient populations and the scarcity of prior research and data. Traditional drug repurposing relies on manually cross-referencing heterogeneous datasets to uncover existing drugs with new therapeutic applications; studies have shown they can uncover drug–disease associations from diverse literature sources.[p59] However, KGs are

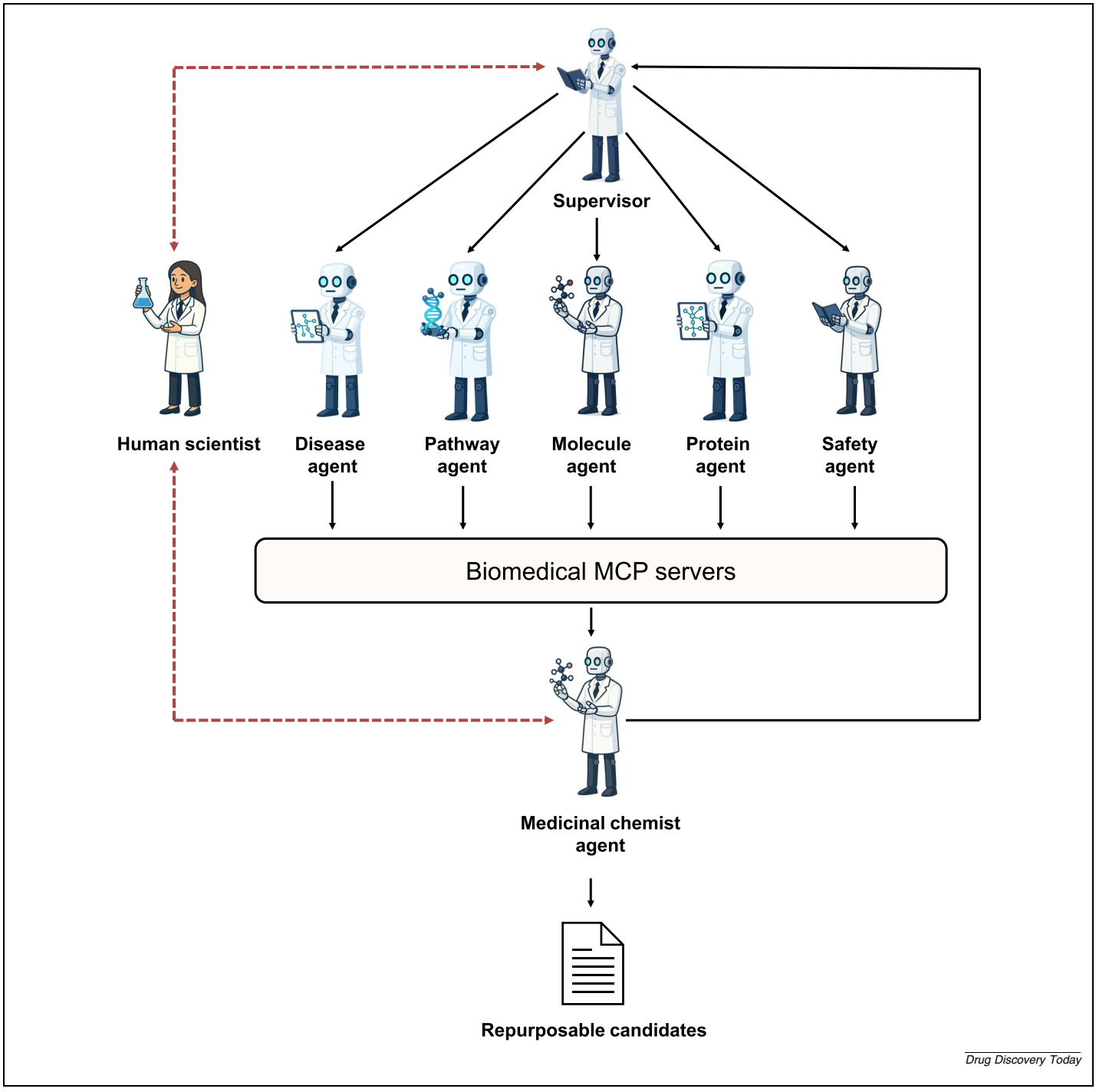


FIGURE 5

Agentic system to accelerate drug repurposing for rare diseases. This system uses the supervisor multi-agent architecture, with five sub-agents, each is responsible to work with one specific domain, including disease agent, pathway agent, molecule agent, protein agent and safety agent. At the termination, there is one medicinal chemist agent that can validate the results. Humans can interact with supervisor and medicinal chemist agents.

static, limited to surfacing associations without the ability to actively reason across data sources, adapt to new evidence or coordinate the multi-step processes required for drug repurposing. To address these limitations, an agentic MCP-driven system was implemented by Augmented Nature with a supervisor coordinating specialized agents, each handling a distinct part of the discovery pipeline (Figure 5). The disease agents retrieved genes and variants from Ensembl[p60] and OpenTargets[p61]: the pathway agent mapped biological processes via Reactome[p31] and KEGG[p62]; the protein agent collected structural data from AlphaFold[p32] and PDB[p63]; the compound agent searched ChEMBL[p28] and PubChem[p29] for molecules targeting those proteins; and the safety agent evaluated toxicity and ADMET profiles to filter out poor candidates. This pipeline ensures only promising molecules progress.

As a case study, this system was used to find repurposing opportunities for spinal muscular atrophy (SMA). The orchestrator launched agents in parallel: the disease agent identified proposed key genes like SMN1 and SMN2; the pathway agent mapped related biological mechanisms; the protein agent retrieved structural data; and compound agents searched for ligands. The safety agent filtered for favorable predicted ADMET properties (Table 4). This automated, parallelized approach greatly reduced manual effort and provided comprehensive data coverage. The system generated a shortlist of repurposable ligands within hours, compared with the weeks required by manual methods. It integrates genomic, pathway, structural and chemical evidence in a single pipeline similarly to Wang and colleagues.[p64] The search often surfaces drugs already approved or in clinical development, potentially lowering regulatory and financial barriers for further testing (of course at the price of potential limitations to IP and economic upside of a project). Importantly, this approach is scalable to any rare disease with minimal setup, making it a potentially powerful tool for accelerating early discovery.

Automating small-molecule synthesis

Synthesis of candidate compounds is often a rate-limiting step in small-molecule discovery,[p65] sometimes taking up to 6 months and costing tens of thousands of dollars. Automating this process can shorten iteration cycles and lower drug development costs. The problem of small-molecule synthesis automation is twofold.[p66] First, most drug candidates lack published synthetic routes. Because there are 10^{60} drug-like compounds available, synthesis planning becomes crucial. Recovery from failed experiments is another aspect – a reaction mixture might be rescued by changing reagents, solvents, temperature or routes. Second, automating synthesis execution is complex owing to harsh conditions, corrosive compounds and solvents that can damage equipment.

AI agents can address planning synthetic routes and automating execution. Retrosynthesis tools and chemical databases enable agents to retrieve and analyze methods, while robust benchmarks track model performance. Full autonomy requires integration of models with physical experiment execution, which is challenging. Several studies attempted tight integration between experiment execution hardware and LLM-based agents.[p20],[p27],[p67],[p68] Following on from these, onepot.ai combines hardware, including a tube picker, decapper, liquid handler, plate sealer and LC/MS for analysis, with virtual tools for literature search, retrosynthesis and code execution (Figure 6a). The virtual system uses experimental results and broad chemical knowledge, enhancing planning robustness (Figure 6a). Additionally, agents were given access to the hardware layer used for the physical execution of experiments, facilitating transfer from hypothesis generation to experimentation. This dual approach, which combines virtual and physical components, moves closer to a true 'AI scientist' that can generate hypotheses, validate them experimentally and learn from outcomes.

This system was used to design and execute small-molecule synthesis experiments, bringing discovery closer to a closed-loop process. Advances in LLM reasoning and agentic tools further improved planning, analysis and hypothesis generation. The facility enabled automated synthesis across seven reaction types as shown in Figure 6b, with two-to-five protocols per type, providing access to a large pool of high-quality chemical reactivity data that were collected in data collection and production synthesis settings. The former included running experiments with a variety of conditions and diverse combinations of substrates. These data were used to develop ML models that achieved

TABLE 4

Agent workflow and quantitative outputs for SMA drug repurposing.

Agent type	Data sources via MCP	Query parameters	Quantitative outputs	Processing time
Disease agent	OpenTargets v24.09, Ensembl Release 112	Disease: EFO_0008525; association score $\geq$0.70; evidence sources $\geq$3	Two primary genes (SMN1, SMN2); evidence scores: 0.72–0.81	~5 min
Pathway agent	Reactome v88, KEGG 2024-Q4	Gene: ENSG00000172062; enrichment $P < 0.01$	snRNP assembly, RNA splicing, motor neuron pathways	~5 min
Protein agent	AlphaFold v4, UniProt 2024_05	Protein: Q16637; structure confidence pLDDT >70	SMN protein structure, HSPB1/HSP27 target identification	~10 min
Compound agent	ChEMBL v34, PubMed	Target: HSPB1; activity pChEMBL $\geq$6 (IC50 $\leq$1 μM); years: 2022–2025	188 bioactive compounds; 20 publications (5.8% precision from 347 results)	~5 min
Safety agent	DrugBank, FDA databases	Status: FDA approved OR Phase II+ OR SMA literature evidence	47 clinically feasible candidates (25% retention from 188)	~5 min
Complete pipeline	**Eight integrated databases**	**Multi-stage filtering cascade**	**Ten final ranked candidates (4 Tier 1, 3 Tier 2, 3 Tier 3)**	**~30 min total**

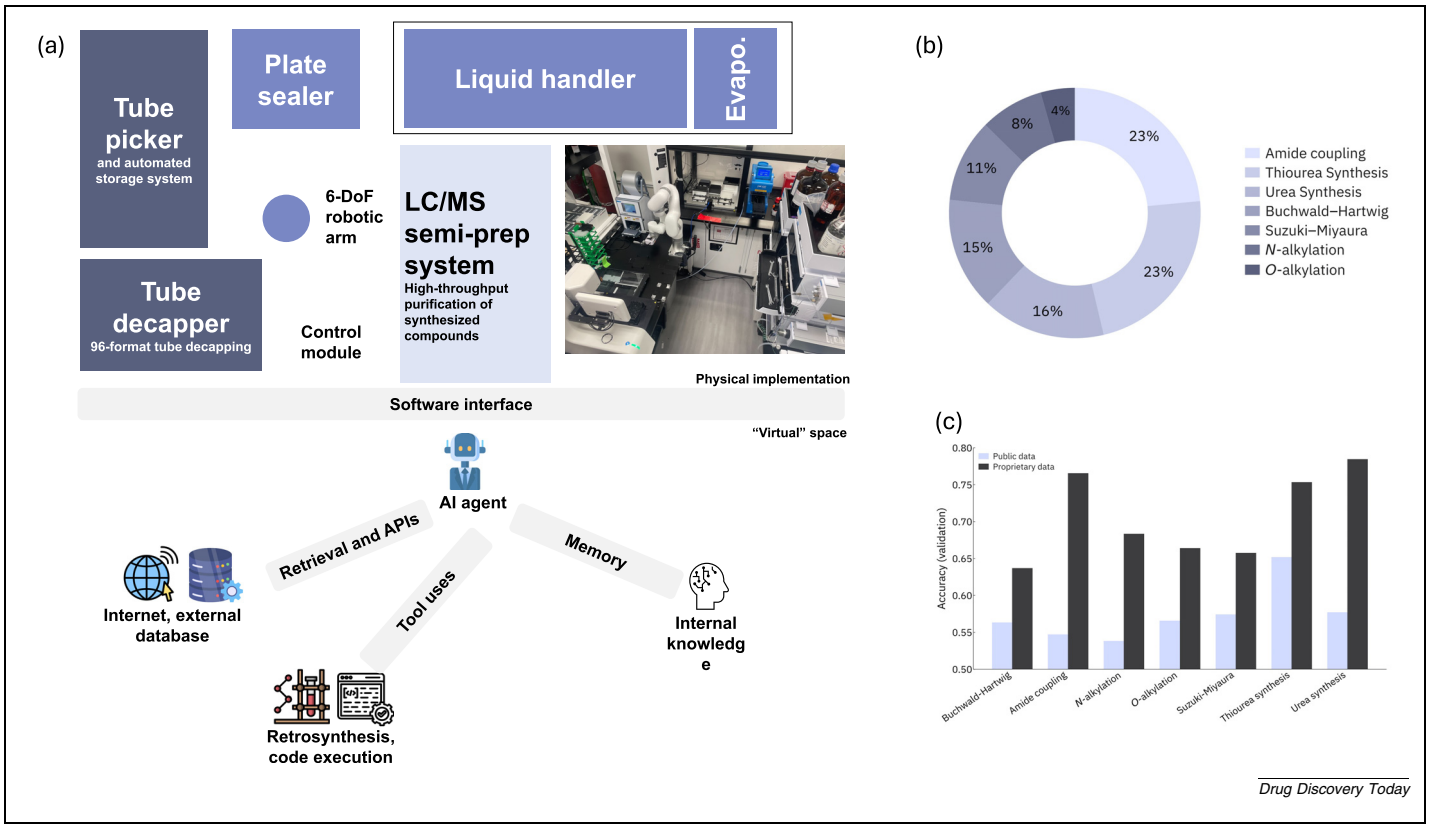


FIGURE 6

(a) Automating small-molecule systems with hardware automation and AI. An agentic system for organic chemistry experimentation is split into physical implementation and virtual space. Automated synthesis platform data and machine learning (ML) performance across reaction types. **(b)** Distribution of experiments performed across seven reaction classes in the automated synthesis facility. **(c)** Validation accuracy of ML models trained on public versus proprietary experimental data for each reaction type.

accuracy values between 63.8% and 78.5% (Figure 6c). Throughput allowed the synthesis of tens of compounds per day with two-step reactions. Agentic platforms can significantly accelerate the iteration cycles that have been traditionally bottlenecked by the 'make' step. Molecules that can be synthesized with this platform can be filtered for properties such as drug-likeness or other properties relevant for a particular use-case.[p69]

Augmenting drug discovery workflows with better search via focal graphs

Large-scale biomedical datasets are complex, diverse, noisy, sparse, siloed and often poorly annotated. Graph-based approaches, such as PageRank,[p70] and crowd sourcing are well-suited for retrieving information from such datasets, making them an attractive foundation for AI agents. Internet searches, which focus on interlinked text, are already used in RAG-based applications. Here, focal graph approaches generalize this to abundant, minimally processed non-textual data.[p71] Comprehensive KGs are useful for network analysis but direct analysis is computationally expensive and visualization unwieldy. Focal graphs are extracted portions of larger graphs relevant to a specific query. By focusing on the most relevant data points, they simplify large dataset exploration, preserve meaningful relationships and reduce computational requirements (Figure 7). When using diverse query types, such as gene expression, chemical structures and cell lines, entities strongly associated with the queries become central to the focal graph, exposing specific data connections and making results concise and transparent.

Focal graph searches have generated data-driven hypotheses in applications like uncovering mechanisms in Cell Painting datasets,[p72] discovering molecular targets for compounds and identifying off-target toxicity mechanisms based on chemical structure and transcriptomics data.[p73] As a case study, authors affiliated with Plex Research used focal graph-enabled LLMs to conduct complex, extended, *in silico* research programs that produce novel, verifiable, data-driven hypotheses. A focal graph-enabled AI agent, prompted with: 'Please plan and execute a research program to identify a novel oncology target in the Wnt pathway', proceeded to list well-known members of the Wnt pathway, identified RNA seq profiles where these pathway members were perturbed and then ran focal graph searches to identify other, potentially novel genes that give rise to similar gene expression profiles when perturbed. This systematic 'phenocopier' approach led to the identification of several potentially novel oncology targets in the Wnt pathway, including the eIF2 complex.[p71] Overall, the workflow shows that *in silico* efforts using AI agents can be augmented with the ability to search data across essentially any therapeutic area, treatment modality, data type or pipeline stage.

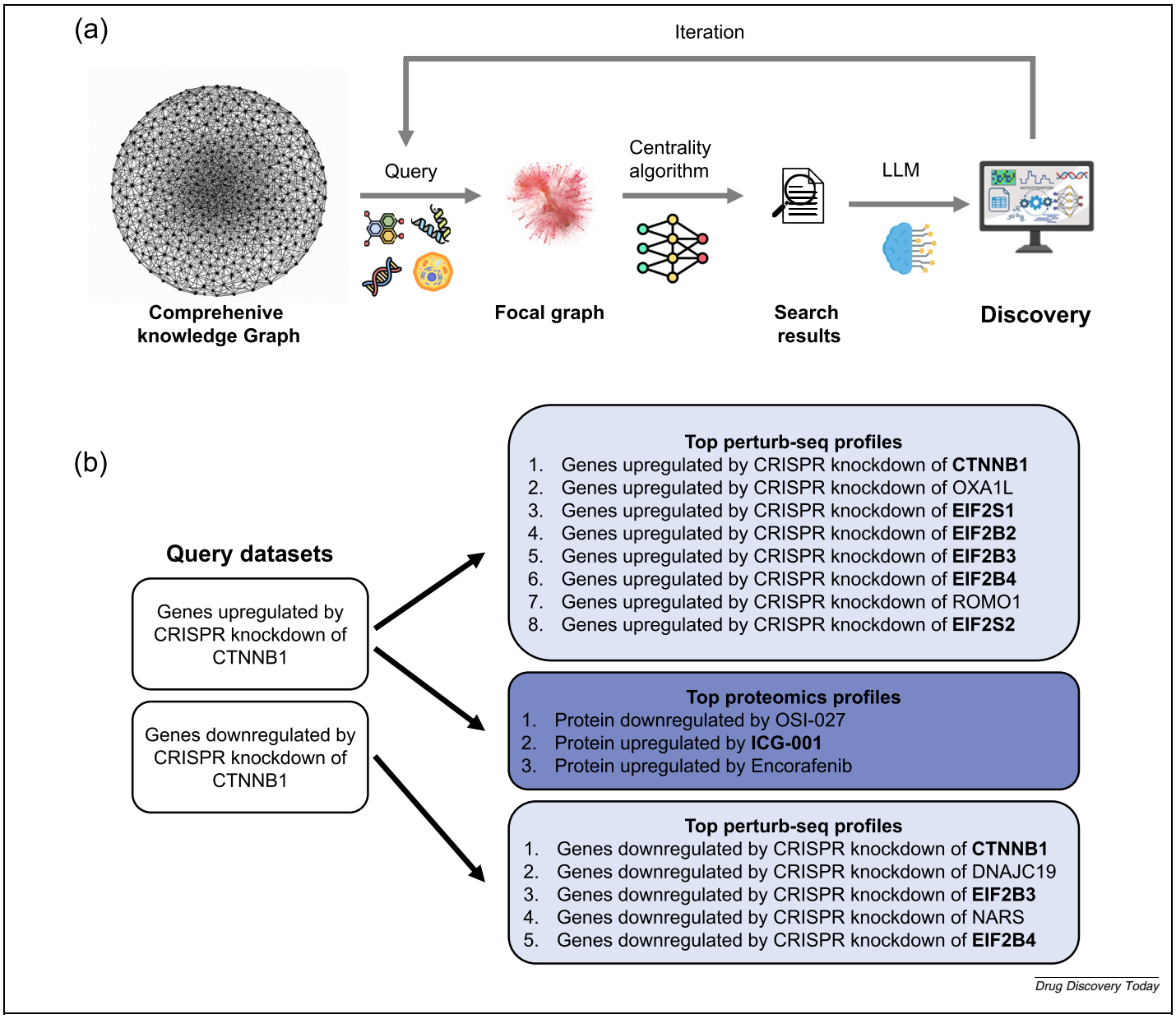


FIGURE 7

Workflow that uses focal graph as data source for large language model (LLM) reasoning. **(a)** The focal graph is extracted from a comprehensive KG, which then applies a centrality algorithm to provide informational data that supports LLM reasoning. **(b)** Discovery of novel components in the β-catenin pathway by focal graph analyses of genes upregulated or downregulated after CRISPR knockdown of β-catenin53.

Discovery-to-deal decisions

Biopharmaceutical asset discovery is essential for pharmaceutical companies, because it involves identifying and evaluating new drugs, technologies or IP for licensing or acquisition. Early-stage biopharmaceutical asset discovery is hindered by manual, fragmented processes and crucial data dispersed across unstructured sources. Conventional tools struggle to connect assets owing to transliteration and alias drift, and integrating scientific, commercial, operational and regulatory information is inconsistent. A multi-agent system can automate end-to-end asset discovery, evaluation and strategic packaging, synthesizing these domains efficiently.

Authors at Convexia Bio developed an agentic workflow for small-molecule asset search and evaluation (Figure 8). The system uses a feedback loop for continuous refinement across five modules: (i) asset discovery, which retrieves and integrates evidence into a query-specific KG; (ii) scientific evaluation, running simulations to assess drug properties; (iii) market analysis, generating market, regulatory and IP scenarios; (iv) clinical assessment, scoring risk using historical trial data; and (v) probability-of-success scoring, predicting clinical trial outcomes. Users specify parameters such as target, geography, modality, development stage and delivery route. The system retrieves and validates data from multiple sources, storing evidence in a

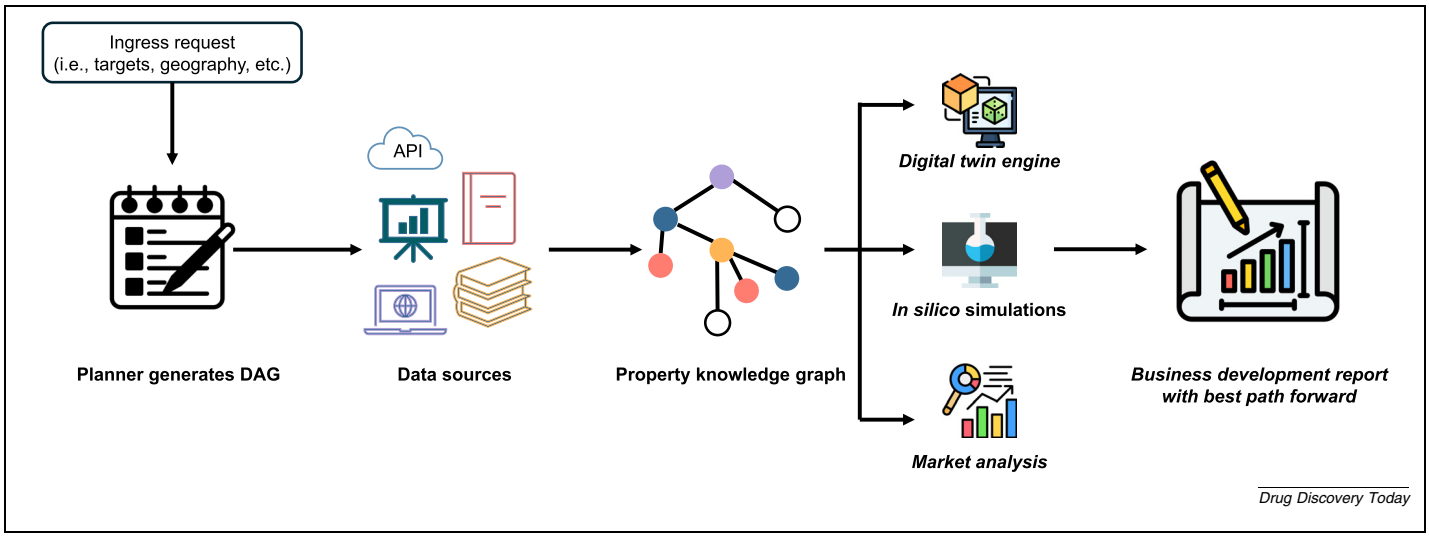


FIGURE 8

Multi-agentic system for discovery-to-deal decision. The planner constructs a task that orchestrates data ingestion from data sources into a per-query property knowledge graph (PKG). The PKG underpins scientific, clinical and strategic modules enabling digital-twin simulations, *in silico* analyses and market modeling. Insights are synthesized into a business development report recommending the optimal licensing or partnership path forward.

centralized KG accessible by all modules. Technical evaluation combines physics-based simulations with ML models. When predictions disagree, uncertainty is flagged, and further validation might be suggested (Table S3, see supplementary material online). The clinical module predicts risks using historical trials and real-world data, whereas market and business modules analyze claims, regulatory filings and pricing databases. Financial projections consider patient population, pricing and market penetration. The probability-of-success module scores each asset's likelihood of clinical success using outputs from the previous diligence agents (Table S4, see supplementary material online). All analyses, assumptions and data sources are documented for transparency and informed go/no-go decisions.

A mid-cap pharma company used the system to triage its portfolio and select candidates for advancement. The system assessed delivery evidence, efficacy, safety, business strategy and the IP landscape. Region-tuned harvesters processed diverse sources to populate a property knowledge graph (PKG). Scientific and clinical models quantified effects and risks, whereas strategic modules mapped commercial and licensing pathways. The business development agent identified target partners. The result was a PKG-backed shortlist with side-by-side comparisons and an outreach plan, delivered in hours at lower cost. The system outperformed manual workflows owing to its broad data coverage and agentic approach.

Challenges and future direction of agentic AI in drug discovery

Agentic AI in drug discovery faces challenges that are not only technical but also scientific, regulatory and operational in pharmaceutical R&D. In this section, we explore these challenges and discuss future directions and opportunities to enhance the integration of agentic AI technologies in drug discovery.

Challenges

Heterogeneous chemical and biological data

The core challenge of using agentic AI for drug discovery is the heterogeneous nature of chemical and biological data.[p74] Table S5 (see supplementary material online) indicates several characteristics of data used in the drug discovery process and their impact on the agentic AI system. Unlike data in image processing or speech recognition, which can be considered independent, data in drug discovery are highly conditional.[p75],[p76],[p77] For instance, the effectiveness of a single compound could depend not only on its molecular structure but also on factors such as assay setup, concentration, genotype and patient characteristics[p76]; this complexity presents challenges at the tool level for AI agents, such as the underspecification of 'tasks', which do not fully represent the real-world situation.[p78] Furthermore, different representations of chemical and biological entities create a system-level challenge: incompatibility. In the chemical domain, a single molecule can be represented in various ways that do not directly translate with each other, including SMILES strings, IUPAC names, InChI codes and database-specific identifiers[p79] such as ChEMBL,[p28] PubChem[p29] or DrugBank[p80] accession numbers. This variety of identifiers that lack a 1:1 correspondence complicates the task of unifying and interpreting chemical knowledge across sources, which is one of the primary purposes of building an agentic AI system. The challenge becomes further complicated in the biological domain, where entities exist at various levels of organization, including genes, proteins, cells, tissues and species.[p76] Engineering effort to map and standardize ontologies is, hence, indispensable for agentic AI systems that are designed to work across diverse biomedical databases.

Privacy and security concerns

Drug discovery is uniquely constrained by IP protection, regulatory documentation and proprietary datasets. Therefore, the use of highly autonomous agentic AI systems must be carefully

evaluated, because they pose acute privacy concerns owing to their capacity to autonomously access, manipulate and examine sensitive data autonomously, including confidential experimental results.[p81] These risks increase when such systems are built upon closed-source LLMs, where the opacity of internal architectures prevents independent verification of data handling practices. The lack of auditability of such models over inadvertent memorization of sensitive information, unrecorded retention or unauthorized cross-domain inference exacerbates exposures and undermines compliance against stringent frameworks.

Another major concern in AI security is the prompt injection attack. In this technique, attackers manipulate LLMs to ignore their existing safety guardrails and generate content they would normally be restricted from generating.[p82],[p83] This issue is even more severe with the advent of agentic AI systems, because they could be fooled into performing actions they were forbidden to execute. A prompt injection can be as simple as a line of text, like: 'Ignore all previous instructions and do [action]'. Such an action could be malicious data collection, exfiltrating confidential molecular structures or trial data if safeguards are insufficient.

Hallucinations

The non-deterministic property of an agentic AI system brings flexibility but also risk. This is because the flexibility of agentic AI systems relies on LLM reasoning capabilities, meaning that they still carry a certain risk of hallucination.[p84],[p85] In low-stakes domains, hallucinations might be tolerable. However, in drug discovery, they can waste months of experimental effort. The examples of hallucination risks in drug discovery include proposing synthetically infeasible compounds, citing non-existent clinical trials to prove a drug's efficacy, misinterpreting correlative omics signatures as causal mechanisms and recommending unsafe *in vivo* dose escalation strategies. The hallucination risk becomes more significant when the AI agent can interact with the real-world environment, such as a scenario where an agentic AI system has access to a lab automation robot; it could potentially conduct misconfigured, redundant or even dangerous experiments owing to the hallucinated coordination from the LLMs. To address this issue, human-in-the-loop should be enforced and processing transparency should be maintained. The human-in-the-loop system not only allows the users to declare permissions on what the agent can perform but also how the users themselves can modify those activities and cancel them instantly if necessary.

Current benchmarking focus on outcomes, instead of solution trajectories

As an early emerging field, agentic AI systems still lack a proper benchmarking evaluation framework specified for the drug discovery domains, and some evaluations presented seem overoptimistic. To maintain scientific transparency despite the proprietary nature of these tools, the community is increasingly relying on standardized benchmarks. Frameworks like BixBench[p86] enable the objective assessment of agent performance across 296 bioinformatics tasks, providing a baseline for comparison (on a task level) that does not require access to the agent's underlying code but rather evaluates its functional output and reasoning capabilities. A remaining limitation is that it focuses primarily on the agent's outcome rather than the reasoning or tool-calling trajectory that led to that outcome. This is also the case for current general-purpose agentic AI benchmarks.[p87],[p88] Creating benchmarks that evaluate intermediate steps could help develop agents that minimally hallucinate and follow logical paths.[p87] Another limitation of orchestrator performance is that it has not yet been benchmarked against a standardized public dataset for literature extraction accuracy. As community-driven benchmarks such as BixBench[p86] mature, systematic evaluation against shared baselines will be essential to validate the generalizability of these results beyond the case studies presented here.

Even evaluating output can be challenging, often leading to overly optimistic results. For example, an original paper by a Google AI co-scientist showcased three instances in the drug discovery field to highlight the capabilities of the system.[p89] One of these examples focused on drug repurposing, where the AI co-scientist suggested novel candidates for treating acute myeloid leukemia (AML). The article asserted that such candidates were validated to inhibit tumor viability at clinically relevant concentrations in several AML cell lines. It must, however, be noted that inhibition at a cell level might not necessarily translate to success at a tumor level. Various factors come into play, such as tumor permeability, heterogeneity of tumors, drug resistance, as well as the microenvironment of a tumor. In summary, whereas the AI co-scientist has great promise for accelerating drug discovery, its assertions need to be evaluated in the appropriate application context, such as real-world complexity *in vivo*.

Future directions

Recent advancements in agentic AI for drug discovery, including autonomous labs, multi-agent architecture and co-scientist collaboration, indicate a rapidly evolving field with significant potential for further development. Here, we propose a visionary pattern of integration that could transform isolated scientific experiments conducted over time at different labs across the world.

Closing the loop: self-driving drug discovery at scale

A major future theme is the rise of self-driving laboratories that execute closed-loop experimentation at unprecedented speed and scale. Although in some areas of research this might remain impractical in the near future, engineering-oriented applications are closer to putting this concept into practice. Recent prototypes graft LLM controllers onto suites of lab instruments (i.e., robot arms) to autonomously plan and run experiments end-to-end.[p19],[p36],[p90] Such autonomous DMTA loops have the potential to accelerate molecule optimization and bioassay cycles. As these systems mature, we anticipate scaling from single steps or reactions to entire workflows of multi-step synthesis and parallel bioassays, with cloud-connected labs continuously evaluating hypotheses. The vision is automated labs that, given a drug target or hypothesis, can continuously design and execute experiments 24/7, expediting discovery while freeing researchers from routine lab work, enabling them to solve strategic and creative problems.

Virtual laboratories and digital twins

To fully harness agentic AI, drug discovery will increasingly involve merging *in silico* modeling with physical experimentation via digital twins in real time. A digital twin is a virtual representation of a physical system with real-time information exchange between the real world and virtual space.[p91] Future

agentic platforms will probably use digital twins to pre-screen and optimize experiments virtually before committing resources in the wet lab. By leveraging high-fidelity simulations and AI modeling, a twin can explore many 'what-if' scenarios (e.g., molecular variations or assay conditions) and identify promising candidates or conditions for the AI agent to test next. For example, using whole-cell simulations, a field that is also rapidly evolving,[p92] an autonomous biology lab could maintain a digital twin of a cell-line experiment. However, performance evaluation needs to be realistic, whether on the cellular level[p93] or on levels of even more complex biology, and current deep-learning and foundation models have not always lived up to expectations.[p94] Once successfully implemented, hypotheses would first be explored in simulation, with only the most informative experiments executed by robots on real cells, and the resulting data fed back to refine the reasoning process. Such closed-loop digital–physical systems could accelerate autonomous optimization while also yielding insights. Realizing this vision will require progress in realistic simulations and in standard protocols for linking instruments to their digital counterparts.

Human copilots and democratized innovation

The rise of agentic AI does not sideline humans; instead, it shifts their position within the drug discovery pipeline. This advancement allows them to overcome lower- and implementation-level obstacles (for example, in writing code or visualization/evaluation standards) and focus more on high-level inductive reasoning tasks. In this future, autonomous research loops completely handle routine tasks and brute-force searches, freeing scientists to focus on creative design, interpretation and strategic decisions.[p95] Medicinal chemists and biologists will work alongside AI copilots that suggest experiments or compound designs, while the humans apply intuition, ethical judgment and contextual knowledge that AI lacks. This human–AI partnership model is particularly important in chemical risk assessment, where the consequences of autonomous errors, from wasted synthesis resources to safety risks, demand that human domain experts retain decision authority at key junctures, even as routine tasks are increasingly delegated to agents. Moreover, with agentic AI transitioning from theory to reality, we expect new sets of industrial standards and regulation for its application to emerge. Just as Good Laboratory Practice (GLP) ensures quality in manually conducted experiments, analogous guidelines will ensure AI-designed experiments are documented, reproducible and safe. Early policy steps (i.e., EU's AI Act) hint that future autonomous discovery platforms will operate under defined governance and oversight.[p83] In conclusion, humans, not replaced but enabled by AI agents, will fuel the next era of innovation if such tools are widely available.

Concluding remarks

AI agents in drug discovery have shown significant potential and early promise in reducing costs, accelerating timelines and increasing overall success rates in pharmaceutical development. This overview has presented diverse case studies covering applications of agentic AI systems across the drug discovery pipeline, from literature synthesis and toxicity prediction to automated protocol generation and end-to-end decision-making. These systems in some cases demonstrate substantial gains in speed, reproducibility and scalability, compressing workflows that traditionally required months into hours while maintaining scientific traceability. The integration of LLMs with perception, computation and action tools enables autonomous reasoning through complex research workflows, addressing longstanding bottlenecks in cross-disciplinary communication and decision making.

The landscape of AI agents in drug discovery is characterized by a dual-track development model. Whereas foundational research often stems from open-source initiatives, the high-fidelity agents specialized for lead optimization and clinical simulation are frequently developed as proprietary platforms by startups and tech providers. These are typically made available to the drug discovery community through software-as-a-service (SaaS) models, ensuring that the high costs of model maintenance, security and continuous training are sustained. To bridge the gap between proprietary silos, emerging standards such as the MCP are becoming essential, providing a universal interface that allows researchers to integrate specialized commercial agents with their internal, private datasets.

Looking forward, the successful deployment of agentic AI in drug discovery will require addressing important challenges including data heterogeneity, system reliability, privacy concerns, robust benchmarking frameworks and real-world use cases. If and when successful, the future holds self-driving laboratories operating in closed-loop experimentation, digital twins enabling virtual pre-screening before physical experiments and human–AI collaboration where scientists focus on strategic and creative problem-solving while AI handles routine tasks. As the field matures with appropriate governance frameworks and industrial standards, we believe agentic AI systems will not replace human expertise but rather augment it, democratizing innovation toward accelerated, cost-effective drug discovery that ultimately benefits patients worldwide.

Conflicts of interest

Each case study reflects work conducted by the respective organization and remains their intellectual contribution. A.E.C. serves as a scientific advisor to Recursion, SyzOnc and Quiver Bioscience and receives honoraria for occasional scientific visits to pharmaceutical companies. All other authors declare no competing interests.

CRediT authorship contribution statement

Dinh Long Huynh: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Srijit Seal:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Dylan Reid:** Writing – review & editing. **Anne E. Carpenter:** Writing – review & editing. **Andreas Bender:** Writing – review & editing. **Ola Spjuth:** Writing – review & editing, Writing – original draft, Conceptualization.

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Appendix A. Supplementary material

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Data availability

No data was used for the research described in the article.

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